

Zero-burn reduces pollution from rice straw burning and wastewater use in biogas production

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Abstract. Open-burning rice straw contributes to global warming. While straw can produce large quantities of carbon dioxide from open burning, this is not considered net greenhouse gas emissions. These lignocellulosic materials may be of great use, but much rice straw is still simply burned, making it possible to avoid this scenario. Therefore this study utilized rice straw and swine farm wastewater for biogas production. The numerous techniques used to produce valuable biogas and the variety of applications that can be made of them. Alkaline pre-treatments were carried out in batch mode to improve biogas production from ensiled rice straw. Prior to batch anaerobic digestion, rice straw was alkali hydrolyzed at three different low sodium hydroxide concentrations (NaOH). The study's findings showed that, compared to untreated and hot water pretreated samples, the sample that had been pretreated with 2% NaOH produced a higher output of biogas. As a result, rice straw and swine wastewater that had been supplemented with the organic matter was a suitable energy crop for the production of biogas and also to avoid environmental pollution further.

Keywords: Zero-burn, Pollution free, Rice straw, Wastewater, Biogas

1. Introduction

Rice is one of the primary grain crops, and it is responsible for the providing of more than fifty percent of the world's population with their daily calorie requirements. Rice is also one of the most widely consumed foods on the planet. Rice is regarded as one of the most significant crops cultivated for human use (Domínguez-Escribá & Porcar, 2010). The widespread adoption of improved rice varieties was the primary driver behind the nearly twofold increase in paddy rice output during the same time span. Despite the fact that there are only two species of rice, there are over 100,000 different types of rice that are farmed. *Oryza sativa*, which is said to have originated in Asia but is currently grown in numerous countries across the globe, is the most prevalent variety of rice. In spite of the fact that Thailand is one of the major exporters of rice in the world, the country's yield is comparatively low in contrast to the yields of other Asian countries (Akagi et al., 2011).

During the harvesting process, rice straw is produced as a byproduct of the rice production process. All through the harvesting process, the rice straw is separated from the rice grains and either stacked up or scattered out across the field, depending on whether the rice was harvested by hand or using machines (Hibberd et al., 2008). The variety and growth influence the ratio of straw to paddy, which can range anywhere from 0.7 to 1.4. Rice straw is a significant agricultural by-product that is commonly used in Asia. Rice straw is produced on a global scale at a rate of approximately 800 to 1,000 million tons per year, with Asia accounting for approximately 600 to 800 million tons of this total (Singh & Arya, 2021). This trend is expected to continue its rapid ascent as a direct result of the shorter turnaround time necessary for increased rice farming. The

development of game-changing combine harvesters, which reduces the high labor cost involved with manual straw gathering, only satisfies half the battle's needs. Due to the short amount of time available for decomposition in intensive systems that produce two to three crops per year, it is not possible to incorporate straw into the soil for the purpose of fertilization. This lack of incorporation leads to poor soil fertilization properties, which ultimately hinders the establishment of crops (Hanafi et al., 2012).

Rice straw has the potential to be processed and used in agriculture for a variety of purposes, including the improvement of soil quality through the processes of carbonization and composting, the generation of bioenergy, and the creation of materials for industrial uses such as bio-fiber and silica (Momayez et al., 2018). This utilization of rice straw in agriculture has the potential to take place. The applications of rice straw that are currently being used have become unprofitable as a result of advancements in both the economy and technology; as a result, these applications will soon be phased out (Dobermann & Fairhurst, 2002). In regions where rice is the predominant agricultural crop, getting rid of such a vast quantity of straw will be a significant difficulty as a result of this. Straw has the potential to be utilized in a wide range of industries, including those pertaining to the production of pulp and paper, construction, soil incorporation, compost, and fuel, to name a few. However, the fact that there are other resources that are in competition with straw makes it unlikely that it will be used immediately for these reasons.

Rice straw, on the other hand, in its unprocessed form is not a particularly healthy biomass source for bioenergy production (Kannah et al., 2019). Pretreatment is an important tool for cellulose conversion processes, and is essential to change the structure of cellulosic biomass to make cellulose more available to the enzymes that convert the carbohydrate polymers into fermentable sugars (Ramaraj et al., 2005a). Either it must be subjected to pretreatment (to enhance its digestibility) or it must be supplemented by the inclusion of other components before it can be considered an adequate feed for digesters (Ramaraj et al., 2005b; Unpaprom et al., 2021). A greater proportion of straw's dry matter is made up of cellulose and hemicellulose than any other component. Lignin, nitrogen molecules, and ash make up the remainder of the substance (Steinbach et al., 2019). Several cutting-edge methods have been conceived of and developed in order to accomplish these aims through the operation of microorganisms (Dussadee et al., 2016; Wannapokin et al., 2018). Animal wastewater contains a huge variety of microorganisms that can efficiently be utilized for biogas production (Nong et al., 2022). Therefore, this study aimed to produce biogas from rice straw and swine wastewater.

2. Methodology

Rice straw was obtained from the agriculture practical zone at the program in Agronomy, Faculty of Agricultural Production, Maejo University, Chiang Mai, Thailand. The materials were chopped into little pieces no larger than 5 mm, which were subsequently ground. The fresh swine wastewater was collected from the Maejo University sawin farm, Faculty of Animal Science and Technology, and the experiment was conducted at Energy Research Center, on the same university campus.

The pretreatment process was adopted by Dussadee et al. (2017). One kilogram of biomass was soaked in 1, 2, or 3% sodium hydroxide (NaOH), as a pretreatment. This reaction mixture was subjected to alkali hydrolysis at 24, 48 and 72h. Additionally, an untreated rice straw biomass experiment has been conducted to compare pretreatment biogas production. In this study, batch digestion methods were used to evaluate the biogas production potential of unprocessed and processed crop wastes. For the batch experiment, the pretreatment effect on the biogas production

rate was investigated. The inoculum was prepared from the ongoing biogas fermenter in the Energy Research Center.

The experiments were carried out in triplicate using glass bottles with a working volume of 800ml and a 1L capacity. The anaerobic tests contained 200g of fresh swing wastewater, 200g of treated and untreated rice straw, and 80 mL of inoculums. The remaining make up water was double distilled. Using the Standard Methods, the solid components, including total solids (TS) and volatile solids (VS), were characterized (APHA, 2005). All pH measurements were performed using a digital pH meter. Using the AOAC official method, the contents of total fat, ash, moisture, fiber, and volatile fatty acids (VFA), cellulose, hemicellulose, lignin, carbon, hydrogen, oxygen, sulphur and nitrogen were measured (AOAC, 2012). Using a biogas analyzer, biogas composition (CH_4 , CO_2 , H_2S , H_2 , and O_2) was determined (GFM 416 series, UK). Analysis using statistics: The mean and standard deviation (SD) of three replicates were used to express the data on the chemical and elemental composition.

3. Results and discussion

3.1. Agricultural crops burning effect

Agricultural burning is used by farmers to remove crop residues after harvesting crops like maize, rice, hay and tobacco stalks (Sophaorn et al., 2022a). Agricultural burning is also used by farmers to remove prunings and trees from orchards and vineyards. Burning can also be used to control pests and prevent diseases, remove weeds, and reduce the risk of disease. Rice crop residue burning has been contributing to major air pollution in many countries. The problems caused by excessive release of particulate matter have also been discussed (Akagi et al., 2011). A concerning deterioration of air quality has been recorded in recent years. The presence of pollutants in the atmosphere, both particulate and gaseous, is a significant factor in the global chemistry of our atmosphere. Open-burning of rice straw in the field is of incomplete combustion in nature; hence, a large amount of pollutants are emitted. Intensive burning of agricultural wastes in many Asian countries may substantially contribute to the formation of atmospheric brown cloud that affects local air quality, atmospheric visibility, and Earth's climate (Sophaorn et al., 2022b).

Open-burning of straw residues also contributes to global warming through emissions of greenhouse gases (GHGs) such as carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O). In addition to this, biomass fires are a significant contributor to the presence of pollutants in the atmosphere. It is a common practice in many parts of the world to do this after harvesting crops in order to get rid of the agricultural leftovers (Sophaorn et al., 2022a). This method has been in use for a considerable amount of time. The main adverse effects of crop residue burning include the emission of GHGs that contributes to the global warming, increased levels of particulate matter (PM) and smog that cause health hazards, loss of biodiversity of agricultural lands, and the deterioration of soil fertility. Stubble burning emits toxic pollutants in the atmosphere containing harmful gases like carbon monoxide (CO), CH_4 , carcinogenic polycyclic aromatic hydrocarbons, volatile organic compounds (VOC). To avoid the air pollution and environmental adverse effect the best way to utilize the rice straw for bioenergy production (Sophaorn et al., 2022b).

3.2. Characteristics of rice straw biomass and pretreatment process

The presence of pollutants in the atmosphere, both particulate and gaseous, is a significant factor in the global chemistry of our atmosphere. In addition to this, biomass fires are a significant contributor to the presence of pollutants in the atmosphere (Dobermann & Fairhurst, 2002). It is

a common practice in many parts of the world to do this after harvesting crops in order to get rid of the agricultural leftovers. This method has been in use for a considerable amount of time. Rice straw is made up of both structural and non-structural components, such as extractives and inorganic ash. These components make up the structure of the rice straw. Cellulose, hemicelluloses, and lignin are the components that make up the structure of the plant (Domínguez-Escribá & Porcar, 2010). Moisture, volatile solids, and ash are of primary concern for proximate analysis; nevertheless, elemental carbon, hydrogen, and nitrogen contents are significant considerations for final analysis. proximate analysis focuses on moisture, volatile solids, and ash. The presence of moisture is the primary focus of proximate analysis, whereas the presence of elemental carbon, hydrogen, and nitrogen is the primary focus of ultimate analysis. The rice straw properties were presented in Table 1.

Table 1. Characteristics of rice straw

Properties	
Cellulose %	35.08
Hemicellulose %	25.90
Lignin %	14.43
Proximate analysis (wt% on wet basis)	
Moisture	12.2
Fixed carbon	14.33
Volatile matter	58.03
Ash	15.48
Ultimate analysis (wt% on dry basis)	
Carbon	40.03
Hydrogen	5.07
Oxygen	37.03
Sulphur	0.12
Nitrogen	0.51

In this study, rice straw underwent an alkaline pretreatment before being delignified to separate the lignin from the cellulose. This was done in order to get the desired result. This activity was carried out as a component of the manufacturing process for cellulose. The bioconversion process makes it possible to get rid of waste products from agricultural production in a way that is efficient, does not involve any risks, and can be relied on (Ramaraj & Unpaprom, 2016). This is accomplished through the fermentation of rice straw that takes place on solid substrates under optimal conditions. In addition to this, it possesses the potential to convert lignocellulosic wastes into usable forms, which can afterwards be utilized in the production of biogas (Wannapokin et al., 2017). Lignocellulosic structures and aerobically stabilised biological sludge represent significant barriers to hydrolysis, which can be overcome by pre-treatment-induced solubilisation. Other particulates, such as those that are larger than hydrolysis-limited, can be reduced with specific pre-treatments (Pantawong et al., 2015; Unpaprom et al., 2015). Substrates may also contain non-biodegradable organic substances, which must be chemically converted in order to become biogas. These traits are present in different substrates, and even within the same category such as plant material or excess sludge the potential effects of pretreatments can vary greatly (Van Tran et al., 2020). Pretreatment of anaerobic digest substrates can improve the process performance by increasing methane yield or reducing solids.

3.3. Anerobic digestion

Pretreatment effect of biogas yield shown in Table 2. Anaerobic digesting is the term for bacteria that works in the absence oxygen to reduce organic matter like animal manure, biosolids and wastewater treatment waste. Anaerobic digestion, which is required for the production biogas, occurs in a closed container known to as a reactor (Souvannasouk et al., 2021). Also, the design and build reactors in many sizes and shapes depending on the location and feedstock associate with anaerobic digestion technology and design. Complex microbial communities in these reactors can break down the waste and digest it. The digestate and the biogas that results from anaerobic digestion are expelled from the digester (Wannapokin et al., 2017). Co-digestion refers to the combination of several organic materials in one digester.

Table 2. Pretreatment effect of biogas yield

Conditions	Pretreatment time (hr)	Added wastewater	Biogas yield of per gram of TS (mL)
Untreated	0	✓	414.15
1% NaOH	24	✓	417.94
2% NaOH		✓	430.85
3% NaOH		✓	453.19
1% NaOH	48	✓	446.19
2% NaOH		✓	492.69
3% NaOH		✓	480.92
1% NaOH	72	✓	486.00
2% NaOH		✓	467.12
3% NaOH		✓	345.08

Manure; food waste, including materials from processing, distribution, and consumption; energy crops; crop residues; fats and oils from restaurant grease traps and many other sources are all examples of co-digested organic materials. Use of co-digestion can help increase biogas production from organic waste with low yields or difficult to digest (Van Tran et al., 2019). Both untreated and treated rice straw was used in the creation of biogas. The wastewater from pig farms was also used. The highest biogas yield and methane production of all the assays was achieved by 2% NaOH pretreatment with 48 hours incubation (Table 3). Passos et al. (2016) reported increases on methane production of 82% with respect to the untreated biomass for alkaline pretreatment, but the methane production from the untreated biomass was very low in this study. Similarly in this study results also confirmed. Therefore, pretreatment is essential step for biogas production.

The variation of the methane yield with respect to the digestion time. Co-digestion could be an effective way to improve the commercial application level of methane fermentation in terms of strong buffer capacity and high methane production at optimal mixing ratios (Ramaraj & Unpaprom, 2016). Co-digestion results show the highest content of methane and the rapid increase in the biogas production, and on the other hand, the low methane content may be due to the presence of some easily digestible from untreated rice straw.

Table 3. Composition of biogas with selected pretreatment

Conditions	Pretreatment time (hr)	Substrates	CH ₄ (%)	CO ₂ (%)	H ₂ S(ppm)	O ₂ (%)
Untreated	0	RS and SW	49.11	50.02	979	0.44
2% NaOH	48	RS and SW	65.5	33.5	641	0.35

4. Conclusion

Rice straw is an excellent substrate for anaerobic digestion, and this is especially true when it is utilized during the process in conjunction with pig manure, as this maximizes the efficiency of the anaerobic reaction. The research revealed that there was a potentially helpful approach for increasing the amount of biogas that was produced, and that method involved mixing the digestion of rice straw and swine manure. The fact that the two different components were co-digested provided some evidence for this theory. Because the utilization of the biomass from weed wastes is essential for the production of biogas, it is essential that the production equipment be efficient, both monetarily and in terms of the impact it has on the surrounding environment.

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